



Week 4 Lecture 4

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	26
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SQL

Relational Databases - SQL

- From IBM ~ 1970s
- Data is stored in tabular format:
 - Columns of tables: fields (name, address, department, ...)
 - Rows of tables: individual entries (student1, student2, ...)
- Key: unique way of accessing a given row
 - **Primary Key**: important for fast access to large databases
 - **Foreign Key**: connect to a different table - Relationships

Queries

- Retrieve data from a database

eg: "Find students with name beginning with A"
"Find all courses offered in 2021"

Structured Query Language

- English-like, but structured
- Quite verbose
- Specific mathematical operations:
 - Inner join

- Outer join

Example: Inner join

<u>Aa</u> Name	<u>≡</u> IDNumber	<u>#</u> hostelID
<u>Sunil Shashi</u>	MAD001	1
<u>Chetana Anantha</u>	MAD002	2
<u>Madhur Prakash</u>	MAD003	2
<u>Nihal Surya</u>	MAD004	3
<u>Shweta Lalita</u>	MAD005	2
<u>Raghu Balwinder</u>	MAD006	3
<u>Gulshan Kuldeep</u>	MAD007	1
<u>Kishan Shrivatsa</u>	MAD008	1
<u>Purnima Sunil</u>	MAD009	2
<u>Nikitha Madhavi</u>	MAD010	1
<u>Lilavati Prabhakar</u>	MAD011	3
<u>Rama Yamuna</u>	MAD012	3

<u>Aa</u> ID	<u>≡</u> Name	<u>#</u> Capacity
<u>1</u>	Jamuna	300
<u>2</u>	Ganga	300
<u>3</u>	Brahmaputra	500

Student - Hostel mapping

```
SELECT Students.name, Hostels.name
FROM Students
INNER JOIN Hostels
ON Students.hostelID = Hostels.ID;
```

```
---- OUTPUT ----
Sunil Shashi, Jamuna
Chetana Anantha, Ganga
```

Cartesian product

- N entries in table 1
- M entries in table 2
- $M \times N$ combinations - filter on them

Powerful SQL queries can be constructed

Example: find all the students in Calculus

- Find ID number for the course
- Look up StudentsCourses table to find all entries with this course ID
- Look up Students to find the names of students with these IDs

```
SELECT s.name
FROM Student s
JOIN StudentsCourses sc ON s.IDNumber = sc.studentID
JOIN Courses c ON c.ID = sc.courseID
WHERE c.name = 'Calculus';
```

Summary

- Models - persistent data storage
- Mechanisms:
 - CSV
 - spreadsheets
 - SQL
 - NoSQL

- Entities and Relationships
 - Different ways of representing

No details on display, views or what kind of updates permitted